

Python Exception Lab — Retry Until Valid Input (0–5)

🎯 Requirements

- User must enter a number **0–5**
- If the input is invalid, the user must be allowed to **re-enter** the number
- Catch:
 - Non-integer input
 - Out-of-range input
 - Divide-by-zero
 - Any unexpected error
- Show the result once valid input is received
- Test it by entering alphabetic characters, a number other than 0 to 5, a 0, and finally a number from 1 to 5.

Solution (with retry loop)

```
# Exception Lab: Divide 100 by a user-entered number (0 to 5)

while True:
    try:
        num = int(input("Enter a number from 0 to 5: "))

        # Check range
        if num < 0 or num > 5:
            raise ValueError("Number must be between 0 and 5.")

        # Perform division
        result = 100 / num
        print(f"100 divided by {num} is {result}")
        break # Exit loop after successful division

    except ValueError as ve:
        print("ValueError:", ve)
        print("Please try again.\n")

    except ZeroDivisionError:
        print("Error: Cannot divide by zero.")
        print("Please try again.\n")

    except Exception as e:
        print("Unexpected error:", e)
        print("Please try again.\n")

print("Program complete.")
```